

CHARTOGRAPHER: Counterfactual Chart Generation for Evaluating Vision-Language Models

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Abstract

Chart question-answering (QA) benchmarks aim to pose questions that require visual reasoning to correctly answer, but models can often reach solutions through shortcuts or prior familiarity with a chart based on their own background knowledge. To strictly evaluate visual reasoning, we propose *counterfactual charts* where the chart-question task remains fixed, but underlying chart and the corresponding answer are varied. We introduce CHARTOGRAPHER, a framework to reverse engineer charts into executable code, validate reconstruction fidelity, generate seed-controlled counterfactual variants, and derive new answers from executable QA logic. We apply this framework to existing chart QA datasets and evaluate proprietary and open-source vision-language models (VLMs), measuring variation sensitivity and generalizability. Counterfactual charts reveal failures hidden by single-chart performance: VLMs often fail to generalize after answering the original chart correctly. We find failures are most prevalent when updated charts require novel visual reasoning pathways.

1 Introduction

Charts are a compact language for communicating quantitative evidence. Answering questions based on information in charts requires more than recognizing visual elements: a model must read labels, identify encodings, make comparisons, aggregate values, track trends, and sometimes combine visual evidence with domain conventions (Kafle et al., 2018; Kahou et al., 2018). As vision-language models (VLMs) are increasingly used to understand and summarize scientific papers, financial reports, dashboards, and other data-rich documents, chart question-answering (QA) has become an important test of multimodal reasoning.

Chart QA benchmarks now cover a wide range of synthetic, web-sourced, and scientific visualizations (Methani et al., 2020; Masry et al., 2022;

Wang et al., 2024). However, they largely evaluate fixed chart-question-answer triples where each question is tied to one specific chart and answer. Benchmarks aim to measure visual reasoning ability, but often the correct answer can be reached through shortcuts, by exploiting regularities in the question, or relying on parametric knowledge acquired from charts or other sources during pre-training. What remains unclear from existing benchmarks is whether a VLM’s behavior on the same chart-question task generalizes when the underlying data changes.

In this paper, we introduce CHARTOGRAPHER, a framework for generating *counterfactual charts* whose answers are recomputed after controlled data changes, to disentangle visual reasoning ability from reliance on shortcuts. The desired behavior is simple: when the visual evidence changes, the answer should change accordingly. We propose chart-question *families* as a primary evaluation unit that includes existing charts, reconstructions, and counterfactual variants. This shifts evaluation from recovering one fixed answer to measuring whether model predictions remain grounded in changed visual evidence. Each family is validated through a base reconstruction that checks whether the task survives chart-to-code reverse engineering before introducing data changes. We then compare behavior within the family to determine whether success on the original chart generalizes to counterfactual variants. Our contributions are:

- We create a chart-to-code pipeline that converts an individual chart QA example into a counterfactual chart-question family, with iterative reconstruction, human-in-the-loop validation, and executable QA regeneration.
- We propose a counterfactual VLM evaluation framework that covers single-chart performance, reconstruction fidelity, variant sensitivity, and counterfactual generalizability.

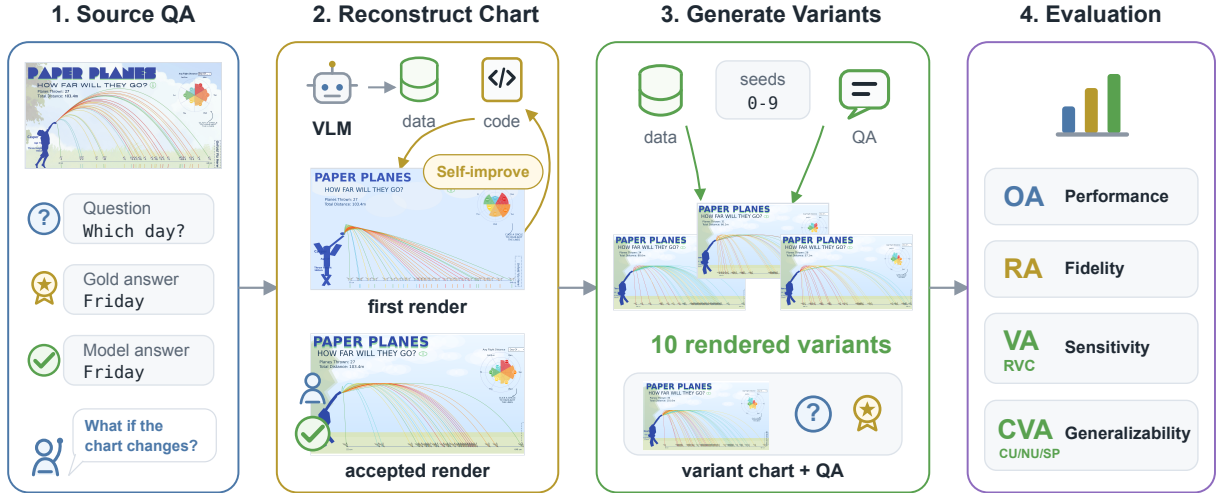


Figure 1: The CHARTOGRAPHER pipeline for constructing counterfactual chart-question families. Starting from a source chart QA example, the pipeline reconstructs semantic chart data and chart-rendering code, iteratively revises the render until accepted, generates seed-controlled counterfactual variants, and recomputes answers with executable QA logic. The resulting families test whether success on an original chart-question task generalizes with altered visual evidence. See Section 3.4 for evaluation metric definitions.

- We apply this framework to existing chart QA benchmarks, showing that success on original charts often fails to generalize when visual evidence changes.

2 Related Work

Chart QA benchmarks. VLM reasoning ability has been studied through both synthetic and human-generated chart QA benchmarks. DVQA (Kafle et al., 2018) and FigureQA (Kahou et al., 2018) use controlled chart or figure generation to isolate basic visual reasoning operations, while PlotQA (Methani et al., 2020) focuses on scientific plots. ChartQA (Masry et al., 2022) combines web-sourced charts with human-authored questions requiring more complex visual and logical reasoning. More recent datasets such as CharXiv (Wang et al., 2024) and ChartMuseum (Tang et al., 2025) move toward realistic scientific figures, human-designed visualizations, and questions that require visual, textual and multimodal understanding. This progression has made chart QA more realistic and challenging, but the dominant evaluation unit remains a fixed chart-question-answer triple. Our work uses ChartQA, CharXiv, and ChartMuseum as source benchmarks, but instead of evaluating only chart-question-answer triples, we convert examples into counterfactual chart-question families to isolate true visual reasoning.

Chart understanding with VLMs. VLMs combine visual encoders with large language models to align visual inputs with text (Alayrac et al.,

2022), and recent frontier systems report broad multimodal capabilities (OpenAI, 2025b; Anthropic, 2025a; Google, 2025; Bai et al., 2025a). Charts remain challenging because models must parse visual encodings, recover quantities, follow labels and legends, and perform numerical operations over the extracted evidence. Prior work aims to improve chart reasoning through intermediate representations, such as screenshot-to-HTML parsing (Lee et al., 2023), chart-to-table derendering (Liu et al., 2023a,b), grounding and reflection (Xu et al., 2025), and distillation of chart reasoning trajectories (He et al., 2025). Our focus is complementary: we use counterfactual chart families for testing whether VLM chart reasoning generalizes when the underlying chart data changes.

Shortcuts and counterfactual evaluation. Shortcut behavior is well documented in language benchmarks: Gururangan et al. (2018) shows that annotation artifacts in natural language inference data can make labels partially predictable without access to the question. The same concern extends to multimodal evaluation, where visual reasoning benchmarks can reward shortcuts rather than application of visual logic, including reliance on parametric knowledge and superficial linguistic regularities (Hou et al., 2025; Chi et al., 2025; Xia et al., 2025). This risk is relevant for charts, where previously encountered images, recognizable templates, and predictable answer distributions may obscure whether a VLM genuinely reasons over visual evidence. Chart-HQA (Chen et al., 2025) introduces

hypothetical assumptions over chart questions to probe counterfactual reasoning over chart content. CharXiv evaluates charts of similar visual complexity with newly annotated questions as an alternative to templated questions (Wang et al., 2024). Our work regenerates the charts themselves with altered data, rather than asking textual hypotheticals over a single chart or replacing charts with visually similar ones, thereby requiring models to reason over changed visual evidence.

3 CHARTOGRAPHER: Counterfactual Chart Framework

Given a chart QA example with a chart image, a question, and an answer, our framework builds a counterfactual chart-question family for the same task, as visualized in Figure 1. The family contains the original chart, a reconstructed chart rendered via reverse engineered plotting code, and counterfactual variants whose answers are recomputed from the underlying data. The reconstruction verifies that the original task survives re-rendering, while the variants test whether predictions remain grounded after alternations to the visual evidence.

3.1 Chart Reconstruction

Chart-to-code reconstruction. The first step reverse engineers a chart image into executable plotting code, drawing on the broader idea of recovering structured data or rendering code from visual inputs (Liu et al., 2023b; Yang et al., 2025). For each source chart, a VLM produces semantic chart data and chart-rendering code: the semantic data records quantities, labels, categories, and groups as seen in the original chart, while the code reproduces visual encodings, layout choices, and rendering logic. This separation is what makes counterfactual generation possible: data values can vary while the chart theme and purpose remain unchanged. Appendix A.2 provides additional reconstruction details.

Self-refinement. Reverse engineering complex charts into data assets and code for re-rendering rarely works perfectly on the first attempt. Hence, we iteratively improve the conversion through a VLM-led self-refinement loop. After the initial data/code pair is generated, the chart is re-rendered and then compared with the original. The VLM diagnoses differences and flaws in the reconstruction, creates a plan for improvement, and then executes that plan to generate another version of data and code. This loop is continued until the VLM raises

no significant concerns, or to a maximum number of iterations (we cap at five). Appendix A.3 provides additional details on self-improvement.

Human-in-the-loop validation. After self-refinement, the VLM classifies its work as acceptable, unsuccessful, or in need of human feedback. Some reconstructions require human judgment because labels are ambiguous, values are unreadable, or the intended visual encoding is not recoverable from pixels alone. Human reviewers inspect difficult but promising cases, approve usable reconstructions, reject low-fidelity outputs, and record assumptions for downstream generation. Appendix A.3 provides additional details on human validation.

3.2 Counterfactual Chart Generation

For each accepted reconstruction, we create a seed-controlled data generator, and use it to generate ten counterfactual charts that resemble the original, but reflect the altered data. The generator preserves the chart schema, rendering constraints, and domain assumptions while changing the data in meaningful ways. Appendix A.4 provides additional details on counterfactual data generation.

3.3 QA Regeneration

Counterfactual charts require valid questions and answers that reflect their altered data. For each accepted reconstruction, we create executable QA logic that computes answers directly from the underlying data, not the visual chart. The original question is kept whenever it remains valid, and is rewritten only when necessary to maintain coherence. This makes large-scale counterfactual labeling feasible without manually annotating every generated chart. Appendix A.5 provides additional details on QA generation.

3.4 Counterfactual Family Evaluation

Each family contains three components: the original chart, the base reconstruction, and counterfactual variants. The base reconstruction is a control: it tests whether the chart-question task survives re-rendering with unaltered data. The counterfactual variants are the intervention: they test whether a VLM remains grounded when visual evidence changes for the same chart-question task. We therefore separate single-chart performance, reconstruction fidelity, variant sensitivity, and counterfactual generalizability:

- **Original accuracy (OA):** Accuracy on the original benchmark distribution.

- **Reconstruction accuracy (RA):** Accuracy on base reconstructions with unaltered data.
- **Variant accuracy (VA):** Average accuracy on counterfactual variants with changed visual evidence and updated questions.
- **Relative variant change (RVC):** Computed as $100 \cdot (VA - OA) / OA$, RVC summarizes sensitivity to variant charts.
- **Conditional variant accuracy (CVA):** Counterfactual generalizability measured as VA restricted to families where the model answered the original chart correctly.

Appendix A.8 defines the metrics and aggregation procedure formally.

To diagnose failed generalization, we analyze variants where the model answered the original chart correctly. Within this conditional subset, each prediction falls into one of three update outcomes: **correct update (CU)**, where the model answers the variant correctly; **stale prediction (SP)**, where it repeats its original prediction and is wrong; and **noisy update (NU)**, where it changes its answer but remains incorrect. We pair these metrics with failure case studies by inspecting charts, variants, labels, and model predictions from persistently difficult families. This analysis highlights recurring bottlenecks such as dense or crowded layouts, multi-step and spatial comparisons, trajectory tracking, value thresholding, and binding labels, legends, or symbols to visual marks; Section 5.5 and Appendix B.6 provide representative examples.

4 Experiment Setup

Datasets. We use evaluation splits from three chart QA sources: the ChartQA validation split (Masry et al., 2022), the CharXiv validation split (Wang et al., 2024), and the ChartMuseum development split (Tang et al., 2025). These sources provide a varied testbed for chart-question tasks across conventional charts, scientific figures, and human-designed visualizations. We randomly sample 462 chart QA tasks from these splits and exclude charts that are unsuitable for chart-to-code reconstruction, such as cases with ambiguous labels, occluded values, or visual encodings that do not support controlled data edits. After filtering, the source sample contains 440 chart QA tasks across the three datasets. Each task is converted into a counterfactual chart-question families with the model-assisted pipeline described in Section 3. Appendix A.1 reports the per-dataset filtered counts.

Models. We evaluate instruction-tuned VLMs

from both proprietary and open-source model families. The proprietary set includes Claude- (Anthropic, 2025b, 2026), Gemini- (Google, 2025), and GPT-family models (OpenAI, 2024, 2025a, 2026a,b) accessed through their respective APIs. The open-source set is run locally and includes Gemma (Google, 2026), InternVL (Zhu et al., 2025), LLaVA (Li et al., 2024), Pixtral (Agrawal et al., 2024), and Qwen variants (Bai et al., 2025a,b). Appendix A.6 lists the model set and inference settings.

Prompting. All models receive a chart image and a question, with the prompt template fixed across all models and tasks. We evaluate only the extracted final answer, not any rationale or reasoning. Appendix A.7 describes the prompt format and answer extraction.

Evaluation. For original charts, predictions are evaluated against the benchmark labels. For base reconstructions and counterfactual variants, predictions are evaluated against labels produced by the executable QA logic. We use binary accuracy as the base correctness metric, employing an LLM judge to determine whether the prediction is equivalent to the target answer. We report the metrics defined in Section 3.4 and use per-model two-sided paired sign-flip permutation tests over chart-question families for OA-to-VA comparisons. Appendix A.7 provides additional details on the evaluation.

5 Results

Counterfactual chart-question families expose a failure mode that single-chart QA cannot measure: a model may answer the original chart’s question correctly but fail to generalize when visual evidence changes. To examine this failure mode, we first account for reconstruction artifacts by checking that the base reconstructions preserve the original benchmark tasks (RA). We then use counterfactual variants to measure variant sensitivity (RVC), and counterfactual generalizability (CVA). Finally, we analyze how failed generalization concentrates in particular reasoning types, and illustrate these patterns with failure case studies. Table 1 compiles our main results, as discussed below.

5.1 Reconstruction Controls

The group-average rows in Table 1 show that RA is consistently close to or above OA across datasets, with the clearest increase on CharXiv. This pattern likely reflects cleaner base reconstructions that increase resolution, reduce incidental clutter, or

Model	ChartQA				CharXiv				ChartMuseum			
	OA	RA	VA	RVC	OA	RA	VA	RVC	OA	RA	VA	RVC
<i>Proprietary models</i>												
Claude Haiku 4.5	0.907	0.933	0.878	-3.2	0.554	0.615	0.580	+4.8	0.472	0.451	0.357	** -24.3
Claude Sonnet 4.6	0.927	0.933	0.905	-2.3	0.682	0.757	0.704	+3.2	0.690	0.704	0.541	*** -21.6
Gemini 2.5 Flash	0.933	0.933	0.891	** -4.5	0.622	0.682	0.659	+6.0	0.641	0.606	0.493	*** -23.1
Gemini 2.5 Pro	0.947	0.927	0.897	** -5.2	0.689	0.757	0.739	+7.3	0.732	0.704	0.529	*** -27.8
GPT-4.1	0.927	0.900	0.869	** -6.3	0.541	0.574	0.544	+0.6	0.507	0.549	0.392	** -22.6
GPT-4.1 mini	0.880	0.900	0.861	-2.1	0.426	0.628	0.484	+13.7	0.528	0.528	0.446	* -15.5
GPT-4o	0.860	0.887	0.841	-2.2	0.426	0.426	0.394	-7.5	0.345	0.366	0.287	-16.7
GPT-5.4	0.927	0.947	0.916	-1.2	0.777	0.764	0.805	+3.7	0.704	0.718	0.554	*** -21.3
GPT-5.4 mini	0.940	0.940	0.917	-2.5	0.689	0.709	0.702	+1.9	0.486	0.514	0.425	-12.5
Proprietary average	0.916	0.922	0.886	-3.3	0.601	0.657	0.623	+3.7	0.567	0.571	0.447	-20.6
<i>Open-source models</i>												
Gemma 4 E4B IT	0.860	0.853	0.839	-2.5	0.358	0.432	0.414	+15.7	0.289	0.261	0.251	-12.9
InternVL3-8B	0.847	0.853	0.816	-3.6	0.318	0.338	0.308	-3.0	0.268	0.239	0.223	-16.6
LLaVA-OneVision-Qwen2-7B	0.707	0.680	0.612	** -13.4	0.270	0.203	0.197	* -27.0	0.190	0.211	0.164	-13.7
Pixtral-12B	0.827	0.787	0.767	* -7.3	0.345	0.392	0.324	-6.1	0.246	0.254	0.211	-14.6
Qwen2.5-VL-7B-Instruct	0.840	0.880	0.790	-6.0	0.311	0.345	0.352	+13.3	0.275	0.211	0.223	-18.7
Qwen3-VL-8B-Instruct	0.907	0.907	0.882	-2.7	0.486	0.534	0.510	+4.9	0.380	0.423	0.339	-10.7
Open-source average	0.831	0.827	0.784	-5.9	0.348	0.374	0.351	-0.4	0.275	0.267	0.235	-14.5

Table 1: Main accuracy metrics for each model and dataset. Metrics, defined in Section 3.4, summarize base performance, reconstruction fidelity, and variant sensitivity. Average rows report arithmetic means within each model group. Significance stars on RVC mark per-model two-sided paired sign-flip permutation tests for original-to-variant accuracy change over chart-question families: * $p < .05$, ** $p < .01$, *** $p < .001$.

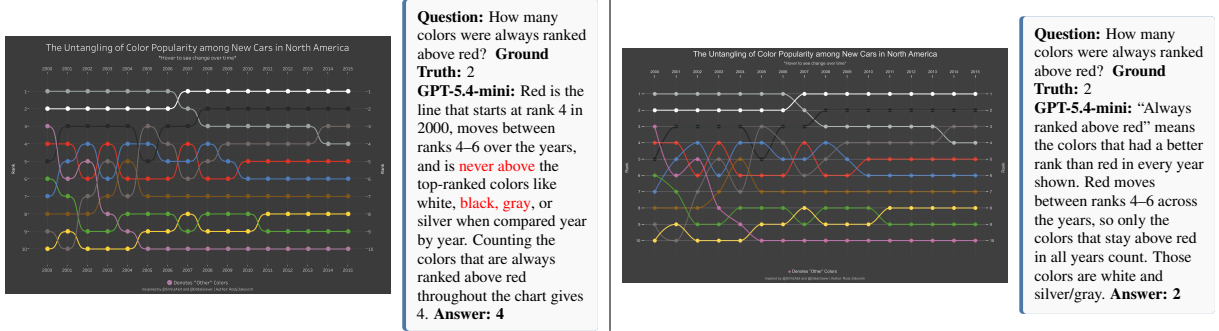


Figure 2: Example where GPT-5.4-mini solves the reconstruction (right), but not the original (left). The reconstruction changes some details such as font sizes, curve shapes, and proportions, and has higher resolution because it was generated by code, not scraped from public sources. Flawed observation shown in **red text**.

simplify graphics, while preserving task-relevant labels and relations. An example where the reconstruction is correctly answered despite the original failing is shown in Figure 2 (see also Figure 7 of Appendix B.1). The two charts have low-level differences such as font choice, colors, and line shape that may contribute to VLM instability. Some original charts in the datasets have low-resolution and visual artifacts because they were scraped from a primary source, whereas our reconstructed charts are generated cleanly at high resolution from reverse-engineered code, which may benefit the VLM’s ability to parse information. However, our main conclusion is that high reconstruction fidelity indicates that the generated charts preserve the benchmark tasks well enough to support counterfactual comparisons.

The per-dataset behaviour diverges once counterfactual data changes are introduced. ChartQA maintains high original and variant accuracy, suggesting that many conventional chart-question families transfer cleanly to the counterfactual setting, even though every models still show a small negative RVC. CharXiv remains difficult in absolute terms, yet its average VA is slightly higher than OA, yielding a small positive RVC that is not reliably different from zero. However, this does not imply that originally solved CharXiv examples reliably generalize. Since VA pools families, gains on originally incorrect examples can offset losses on originally solved ones, which we discuss further in Section 5.2. ChartMuseum is different: RA stays close to OA, but VA drops with RVC showing statistically significant drops in many cases. This

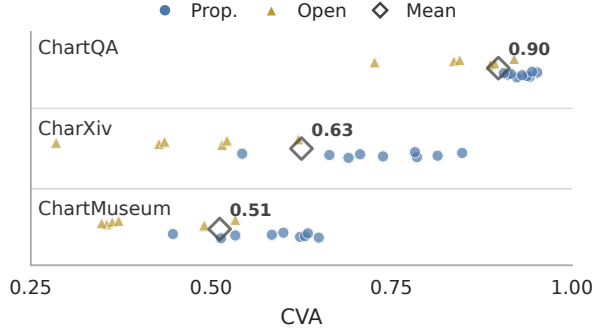


Figure 3: Distribution of conditional variant accuracy (CVA) by dataset and model group. Points are individual models, colors and shapes distinguish proprietary and open-source models, and diamonds mark the mean over model CVA values within each dataset.

indicates VLMs can have trouble handling changed visual evidence distinct from reconstruction artifacts. Open-source models follow the same trends as closed-source, though they are lower in absolute performance across the harder benchmarks.

5.2 CVA Reveals Failed Generalization After Original-Chart Success

The central question we investigate is whether VLMs remain grounded on altered visual evidence after they have demonstrated competence on the original chart. Aggregate VA and RVC are indicative, but do not directly address this. Therefore, we use CVA (Section 3.4) which restricts evaluation to families where the original chart was answered correctly, and then measures Variant Accuracy. This conditioning isolates counterfactual generalizability on solvable questions. Figure 3 reports CVA by dataset and model group, while Appendix B.2 reports the per-model values.

Proprietary models have higher CVA than open-source, indicating better generalization. For datasets, ChartQA has high counterfactual generalizability, whereas CharXiv and ChartMuseum are concerningly low. CharXiv illustrates why the conditional view matters: its average RVC is not reliably negative, but CVA is significantly lower than 1.0. ChartMuseum combines both signals, with a large negative RVC and even lower CVA. Overall, VLMs are not consistently able to generalize to charts with altered visual evidence, even after solving the original. The following sections diagnose potential causes.

5.3 Failed Generalization Reflects Stale and Misgrounded Answers

When visual evidence changes after a successful solve, a VLM may answer the variant correctly

Dataset	Proprietary			Open-source		
	CU	NU	SP	CU	NU	SP
ChartQA	0.929	0.066	0.005	0.851	0.125	0.023
CharXiv	0.730	0.208	0.062	0.468	0.390	0.143
ChartMuseum	0.579	0.327	0.094	0.410	0.432	0.158

Table 2: Update outcomes on counterfactual charts after success on the original, averaged within each model group. For each dataset and group, counterfactual variants are partitioned into correct updates (CU), noisy updates (NU), and stale predictions (SP).

(correct update, CU), repeat its original prediction incorrectly (stale prediction, SP), or change to another wrong answer (noisy update, NU). SP indicates failed re-grounding after the visual evidence changes, which may reflect visual insensitivity, memorization of the original chart, or data contamination. These factors are coupled with reliance on parametric knowledge of the textual question or chart from prior exposure during training. NU can indicate an attempted update which is not grounded in visual evidence, or a reasoning failure. Table 2 displays the outcome rates for these three categories, aggregating over models (Appendix B.3 reports the per-model results).

NU is the more common failure type, while SP is also substantial for the harder datasets like CharXiv and ChartMuseum, especially for open-source models. Thus, models often change their response without computing or grounding the updated answer correctly. The prevalence of SP shows that VLMs sometimes rely on factors other than the presented evidence when answering questions, such as parametric knowledge of the question or chart. This fact has consequences for understanding the relative performance of VLMs on common benchmark datasets like CharXiv and ChartMuseum: strong performance may not indicate true visual-reasoning ability, but merely a leak of evaluation data into the training corpus.

5.4 Generalizability is Weakest for Visually Grounded Questions

We next examine which chart-reasoning demands make counterfactual generalizability difficult. ChartMuseum’s reasoning-type annotations let us test whether failure types concentrate in particular kinds of chart reasoning. Following ChartMuseum terminology (Tang et al., 2025), the categories distinguish Visual, Text, Visual/text, and Synthesis questions, depending on whether the answer is grounded in visual marks, explicit text on

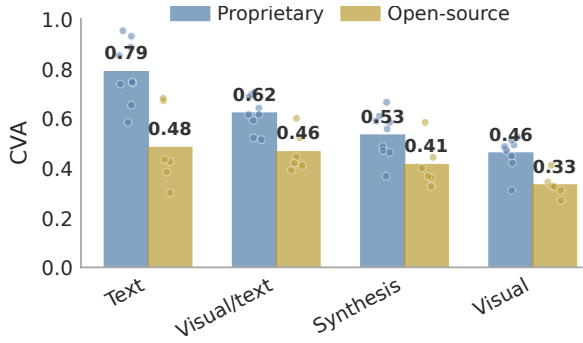


Figure 4: ChartMuseum CVA by reasoning type and model group. Bars show mean CVA for each group, with individual model scores overlaid. Reasoning-type names follow the ChartMuseum terminology.

the chart, either source, or their combination. Figure 4 reports ChartMuseum CVA by category; Appendix B.4 gives the per-model breakdown.

The best generalization occurs when it is possible to rely on text only, followed by Visual/text where text can still be used as a crutch. However, when true visual reasoning is required, as for Synthesis, and Visual cases, generalizability is worse, both for proprietary and open-source groups. Generalizability being weakest for examples that depend on visual grounding further calls into question the extent to which current VLMs are capable of grounded visual reasoning.

5.5 Failure Case Studies

Figure 6 illustrates in particular cases how success on the original chart can lead to a stale prediction, a noisy update, or a correct update after the visual evidence changes with the question fixed. The case-study CU/NU/SP rates are computed over proprietary models only. Motivated by the reasoning-type breakdown from Section 5.4, the cases are selected from ChartMuseum’s Visual family with high stale prediction rate, the Visual/text family with many noisy updates, and the Text family with a high correct update rate for contrast. Appendix B.6 extends the case studies to CharXiv.

The first row shows stale prediction in a visual comparison after the correct answer changes, but the displayed models keep the original prediction. The answer depends on comparing irregular band lengths along the age axis without explicit endpoint indicators, while the variant’s similar overall shape makes the changed relation easy to miss. To further analyse this example, we show reasoning traces from Claude Sonnet 4.6 on the original and variant in Figure 5. The reasoning for the variant is not grounded in the visual evidence, and *better de-*

Original Chart

The Time Wasters category is represented by the red/coral color in the chart. Looking at the red-colored activities, I can identify: Video Games and Watching TV as the main Time Wasters (red color). Watching TV appears to start very early (around age 1–2) and continues throughout the entire lifetime shown (up to age 32) making it the longest duration Time Waster activity. Video Games appears to start around age 8–9 and continues into adulthood but seems to taper off. Watching TV spans from early childhood (age 1–2) all the way through adulthood (age 32), giving it the longest duration among the Time Wasters. **Answer:** Watching TV.

Counterfactual Variant

Time Wasters are represented by the red/coral color in the chart. Looking at the red-colored activities, I can identify: 1. **Watching TV** – appears to start around age 3–4 and continues through to age 32. 2. Video Games – appears from around age 8–9 and continues. 3. Going to Concerts – appears from around age 17–18 and continues. **Watching TV (red) appears to have the longest duration, spanning from very early childhood (around age 3–4) all the way through to adulthood (age 32), making it the longest-running Time Waster activity.** Going to Concerts also appears in red and spans from about age 17 to 32. **Watching TV clearly has the longest duration among Time Wasters, starting in early childhood and continuing through adulthood.** **Answer:** Watching TV.

Figure 5: Claude Sonnet 4.6 responses for the first case study in Figure 6. Highlighted spans show key reasoning points for the original (blue) and their incorrect reuse on the counterfactual variant (red). The reasoning for the counterfactual variant better describes the original chart, a sign of memorization.

scribes the original chart with reuse of key “facts” that appear true for the original but not the variant. This is a clear sign of reliance on parametric knowledge from previous exposure to the original chart and/or QA pair. Hence, reliance on published chart QA benchmarks may conflate visual reasoning ability in VLMs with mere memorization. Instead, our method of using counterfactual chart families generates tasks of a similar difficulty that are certain to not appear in the training corpus of any VLM, giving a disentangled measurement of visual reasoning. Full analysis of reasoning traces for the other models, and further evidence of memorization, in this case study are given in Appendix B.5.

The second row shows a noisy update on an example that requires visual reasoning to locate the peak of a line, and textual reasoning to infer which month this peak occurs in. In contrast to the original, the correct month label does not explicitly appear on the axis and needs to be inferred, making the counterfactual variant more challenging. Only GPT-5.4 makes this update correctly, with other VLMs fixating on the displayed textual information rather than inferring the missing month label. This example demonstrates how counterfactual charts can introduce novel reasoning pathways not required in the original chart.

The third row displays clear party labels, dis-

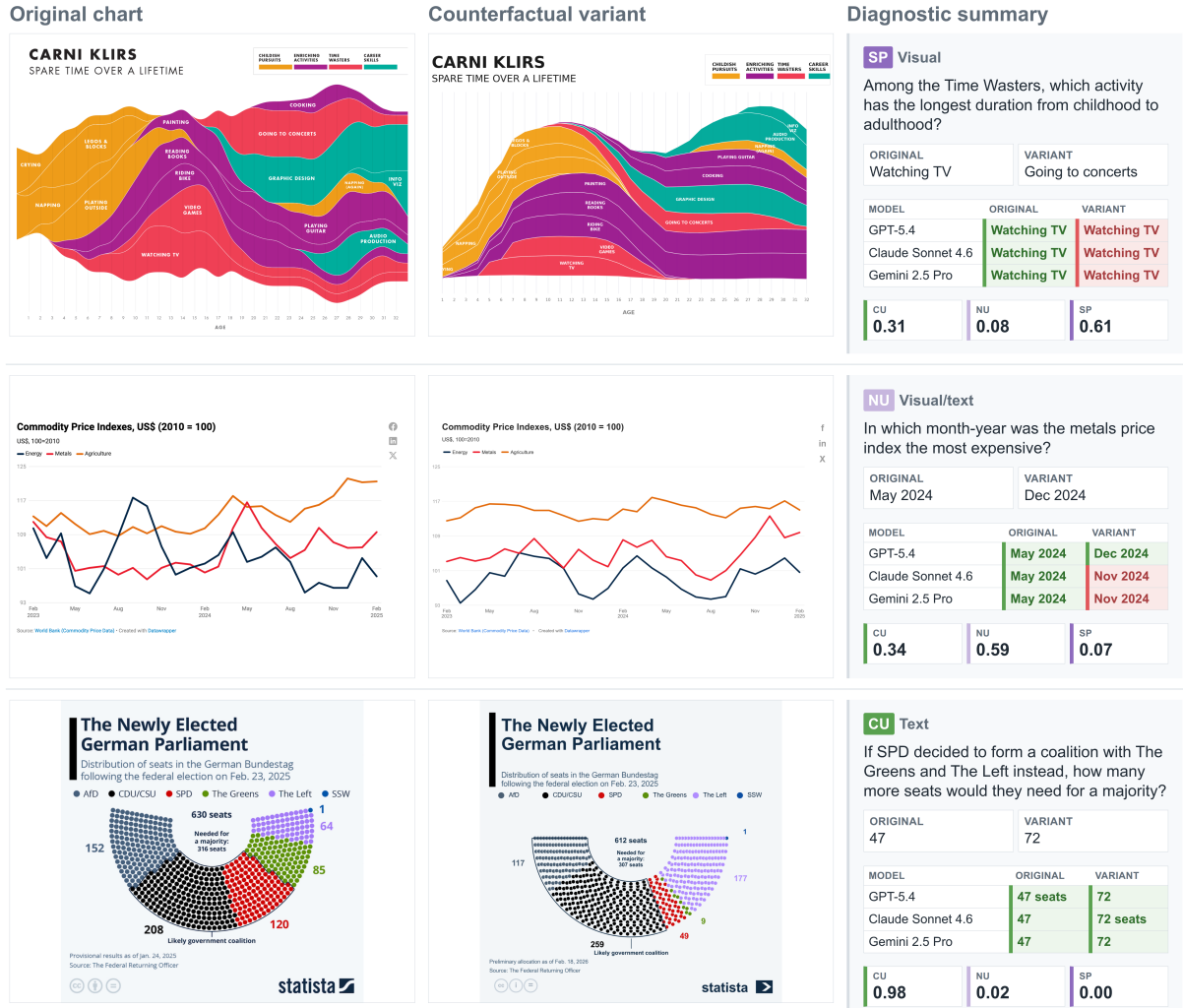


Figure 6: ChartMuseum failure case studies illustrating update outcomes. Each row shows the original chart, one counterfactual variant, gold answers, and representative proprietary-model answers; colors mark correct and incorrect predictions. Summary rates are family-level CU/NU/SP diagnostics over proprietary models and counterfactual variants. Additional reasoning traces for the first two rows are analyzed in Appendix B.5.

tinguishable colors, and explicit seat annotations given in text. Here, models do not need to interpret graphics to reach a solution, but merely extract and associate text correctly before performing a simple calculation. All displayed models correctly update from 47 to 72. Together, the cases illustrate how counterfactual failures can occur after original competence, spanning both stale prediction and noisy updating.

6 Conclusion

We introduced CHARTOGRAPHER, a framework for constructing counterfactual chart-question families that evaluate whether VLM chart QA remains grounded when chart data changes. Our pipeline first reverse engineers code to generate the chart through an iterative process with quality controls, updates the underlying data, and then re-renders

variants that maintain the overall task. Across ChartQA, CharXiv, and ChartMuseum, counterfactual variants expose failures that single-chart evaluation can hide. Models may answer the original chart correctly yet fail the same task when the chart is re-rendered with altered data, with errors split between stale predictions and noisy updates. These failures are especially visible when models must re-ground their reasoning on updated visual evidence, rather than rely on explicit textual cues. Counterfactual evaluation therefore reveals generalizability gaps that single-chart performance can miss, exposes memorization of charts and reliance on parametric knowledge rather than reasoning ability, and overall provides a new tool for diagnosing VLM chart reasoning.

Limitations

Our study has two main limitations. First, the benchmark focuses on charts whose relevant data, visual encodings, and question semantics can be recovered well enough for executable generation, so charts with ambiguous labels or unrecoverable values are filtered out. This makes the counterfactual labels reliable, but it also biases the benchmark toward charts that can be reconstructed and parameterized cleanly. Second, the counterfactual variants preserve the chart-question task while changing the underlying data; they therefore do not test broader shifts in chart type, question style, or visual design. As a result, the benchmark isolates sensitivity to changed visual evidence, but does not measure robustness to changes in how the task is posed or how the chart is encoded.

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A Implementation Details

A.1 Source Data, Filtering, and Licenses

Table 3 summarizes the filtered source samples used in the experiments. We exclude charts when chart-to-code reconstruction cannot support controlled variation, such as charts with distributions that cannot be clearly parameterized or extremely dense values that cannot be reliably recovered from the image. We also exclude cases where the target relation depends on visual details that cannot be reconstructed reliably enough for controlled data edits. The QA-pair count is also the number of accepted chart-question families before counterfactual expansion.

Dataset	Charts	QA pairs
ChartQA validation	75	150
CharXiv validation	148	148
ChartMuseum development	98	142

Table 3: Filtered source samples used for family construction. QA pairs correspond to accepted chart-question families before counterfactual expansion.

We release CHARTOGRAPHER, our counterfactual chart-question families and generation pipeline under the CC BY-SA 4.0 license where permitted by the source terms. Derived examples remain subject to source licenses and provenance requirements: ChartQA is GPL-3.0,¹ ChartMuseum is CC BY-SA 4.0 with chart copyrights retained by original sources,² and CharXiv licenses original non-chart data under CC BY-SA 4.0 while retaining chart copyrights with the source papers.³ We preserve source attribution and release these families for use as an evaluation benchmark.

A.2 Reconstruction

All model-assisted construction calls use a code-capable GPT-family model with temperature set to 0. The reconstruction prompt asks for a JSON object containing semantic chart data and a Python `make_figure(data, savepath=None)` function. The data object contains entities, categories, labels, values, groups, and domain orderings, while style and layout decisions remain in code. The prompt discourages pixel tracing for complex, task-irrelevant visual elements and permits abstraction

that preserves question-relevant evidence.

A.3 Self-Improvement and Human Validation

After reconstruction, automated diagnostics compare the source image and rendered reconstruction. Cases with fix-worthy mismatches enter self-improvement, which can proceed for up to five diagnostic and revision turns before the reconstruction is accepted, rejected, or sent for human review. Reviewers then check layout, marks, scales, labels, legends, panel structure, styling, encoded values, and whether the target question remains answerable from the visible chart. Human intervention is used for recurring problems that automated revision cannot resolve, such as ambiguous or overlapping labels, unreadable values, missing or inconsistent legends, incorrect panel or axis structure, mismatched mark types, or task-relevant relations that automated diagnostics cannot verify. Charts are rejected when the reconstruction is too far from the source image, when the relevant visual relation cannot be recovered, or when the data schema cannot support controlled variation. This review is intentionally task-focused: a reconstruction can be accepted if it preserves the visual evidence needed for the question, even when minor stylistic details differ from the source image. For this work, all human review was conducted by the authors.

A.4 Counterfactual Data Generation

For each accepted chart, a generator prompt produces a deterministic Python `generate_data(data_template, seed)` module. The generator must preserve the template schema, return valid same-schema data, and use seed-controlled regimes rather than only small noise. For each accepted family, the base reconstruction keeps the original reconstructed data, while each seed variant is produced by executing the family-specific data generator with one of ten random seeds, 0–9. The generator keeps the original schema and chart semantics, but changes values, rankings, compositions, or trajectories enough to create a meaningful counterfactual chart. For charts requiring implicit context, assumption files record background knowledge, completeness constraints, and generation guidance.

A.5 QA Generation

For each chart-question pair, the pipeline first produces a Python `adapt_question(data)` module. The adapter normally returns the original

¹<https://github.com/vis-nlp/ChartQA>

²<https://huggingface.co/datasets/lytang/ChartMuseum>

³<https://github.com/princeton-nlp/CharXiv>

question verbatim and makes only minimal validity edits when regenerated same-schema data would otherwise make the question unanswerable. The answer prompt then produces a Python `generate_answer(data)` module that computes the gold answer from the current data object rather than copied constants. The answer module must use fields present in the schema, preserve the answer format implied by the question, and generalize to reordered elements and changed values. Both modules are parsed, restricted to a single top-level function, and executed on the reconstructed data before use.

A.6 Models

We use the same model set for all three datasets. The proprietary group contains Claude Haiku 4.5 and Claude Sonnet 4.6, Gemini 2.5 Flash and Gemini 2.5 Pro, and GPT-4.1, GPT-4.1 mini, GPT-4o, GPT-5.4, and GPT-5.4 mini (Anthropic, 2025b, 2026; Google, 2025; OpenAI, 2024, 2025a, 2026a,b). The open-source group contains Gemma 4 E4B IT, InternVL3-8B, LLaVA-OneVision-Qwen2-7B, Pixtral-12B, Qwen2.5-VL-7B-Instruct, and Qwen3-VL-8B-Instruct (Google, 2026; Zhu et al., 2025; Li et al., 2024; Agrawal et al., 2024; Bai et al., 2025a,b). Proprietary models are queried through their hosted APIs. Open-source models are run locally from Hugging Face checkpoints, using the same prediction and evaluation wrapper as the proprietary systems. Local runs use a single NVIDIA H100 or L40S GPU with 128GB host memory, bfloat16 inference, and a 1024-token generation limit by default. All other inference arguments followed the default settings.

A.7 Prompting and Evaluation

For every original chart, base reconstruction, and accepted counterfactual variant, the model receives only the chart image and the applicable question. Gold answers, reconstructed data, generation seeds, executable code, and regime labels are never provided. The fixed prompt asks the model to answer from the chart, provide a brief rationale, and place the final response inside `<answer>` tags. We score only the extracted final answer; if the tags are missing, we use the model’s final response text. For counterfactual variants, the question is the original question unless the adapter module makes a minimal validity edit.

The answer-comparison judge receives the question, the target answer, and the extracted model

answer, and returns a binary equivalence decision. The judge ignores superficial formatting differences, capitalization differences, and unit formatting when they do not change the answer. Numeric answers are judged by semantic equivalence rather than exact string identity when formatting differs without changing the value. Dates, years, named entities, categorical labels, and ordinal choices must match the target answer. When a question asks for multiple items, the prediction must contain the required set of items without adding contradictory answers. We use GPT-4.1-mini through BespokeLabs Curator as the judge, run with temperature 0, and apply the same judging procedure to original charts, base reconstructions, and counterfactual variants.

A.8 Metric Definitions

For dataset d , let F_d index its accepted source QA pairs; after counterfactual expansion, each $i \in F_d$ defines one chart-question family. Let V_i denote the seed set used to generate variants for index i ; in our experiments, $V_i = \{0, \dots, 9\}$ for all i . For model m and source QA pair i , let y_i^o , y_i^r , and y_{ij}^v be the gold answers for the original chart, base reconstruction, and variant seed j . Let $\hat{y}_{m,i}^o$, $\hat{y}_{m,i}^r$, and $\hat{y}_{m,ij}^v$ be the corresponding model predictions. Here, $\equiv$ denotes judged answer equivalence.

Original accuracy (OA), reconstruction accuracy (RA), and variant accuracy (VA) are

$$\text{OA}_{m,d} = \frac{1}{|F_d|} \sum_{i \in F_d} \mathbf{1}[\hat{y}_{m,i}^o \equiv y_i^o],$$

$$\text{RA}_{m,d} = \frac{1}{|F_d|} \sum_{i \in F_d} \mathbf{1}[\hat{y}_{m,i}^r \equiv y_i^r],$$

$$\text{VA}_{m,d} = \frac{\sum_{i \in F_d} \sum_{j \in V_i} \mathbf{1}[\hat{y}_{m,ij}^v \equiv y_{ij}^v]}{|F_d| |V_i|}.$$

Relative variant change (RVC) is

$$\text{RVC}_{m,d} = \frac{\text{VA}_{m,d} - \text{OA}_{m,d}}{\text{OA}_{m,d}}.$$

Conditional variant accuracy (CVA) restricts VA to families where model m answered the original chart correctly. Let

$$S_{m,d} = \{i \in F_d : \hat{y}_{m,i}^o \equiv y_i^o\}.$$

Then

$$\text{CVA}_{m,d} = \frac{\sum_{i \in S_{m,d}} \sum_{j \in V_i} \mathbf{1}[\hat{y}_{m,ij}^v \equiv y_{ij}^v]}{|S_{m,d}| |V_i|}.$$

This conditioning separates failure to solve the original task from failure to generalize after original-chart success.

Aggregate values reported in the main text and figures are computed from these per-model quantities. For a model group G such as proprietary or open-source models, we report the arithmetic mean over models:

$$\text{Metric}_{G,d} = \frac{1}{|G|} \sum_{m \in G} \text{Metric}_{m,d}.$$

Dataset-level aggregate values use the same unweighted mean over all evaluated models. For example, the aggregate RA/OA pairs and dataset-level CVA values in Section 5 average the corresponding per-model metrics, rather than pooling all model predictions into a single denominator.

Update diagnostics use the same original chart success condition as CVA and include all generated seed variants, because every seed changes the visual evidence even when the answer remains the same. Correctness against gold answers uses the judged answer-equivalence labels. Whether a model changes its prediction is determined by exact match after final-answer extraction and normalization; let $n(\cdot)$ denote this normalized answer string. The restriction is applied separately for each model:

$$D_{m,d} = \{(i, j) : i \in F_d, j \in V_i, \hat{y}_{m,i}^o \equiv y_i^o\}.$$

Because the original-success condition depends on the model prediction, $D_{m,d}$ can differ across models even within the same dataset. Thus, a variant may contribute to the update diagnostics for one model but not for another. With $|V_i| = 10$ seed variants per expanded family, $|D_{m,d}| = |S_{m,d}| |V_i|$. Within each model-specific set $D_{m,d}$, the update rates are

$$\begin{aligned} \text{CU}_{m,d} &= \frac{\sum_{(i,j) \in D_{m,d}} \mathbf{1}[\hat{y}_{m,ij}^v \equiv y_{ij}^v]}{|D_{m,d}|}, \\ \text{SP}_{m,d} &= \frac{\sum_{(i,j) \in D_{m,d}} \mathbf{1} \left[\begin{array}{l} n(\hat{y}_{m,ij}^v) = n(\hat{y}_{m,i}^o) \\ \wedge \hat{y}_{m,ij}^v \neq y_{ij}^v \end{array} \right]}{|D_{m,d}|}, \\ \text{NU}_{m,d} &= \frac{\sum_{(i,j) \in D_{m,d}} \mathbf{1} \left[\begin{array}{l} n(\hat{y}_{m,ij}^v) \neq n(\hat{y}_{m,i}^o) \\ \wedge \hat{y}_{m,ij}^v \neq y_{ij}^v \end{array} \right]}{|D_{m,d}|}. \end{aligned}$$

CU is the fraction of variants where the model predicts the variant answer correctly. SP is the

fraction where the model repeats its original prediction and is wrong on the variant. NU is the fraction where the model changes away from its original prediction but is still wrong on the variant. They form a partition of all seed variants after model-specific original-chart success and sum to one for each model–dataset aggregate. These categories describe observed prediction behavior rather than the causal mechanism behind the error. When update diagnostics are summarized by model group or dataset in the main text, we average the per-model CU, SP, and NU rates over the relevant model set. This keeps each model equally weighted even though the model-specific diagnostic set $D_{m,d}$ can have different size.

B Additional Results

B.1 Reconstruction Accuracy Analysis

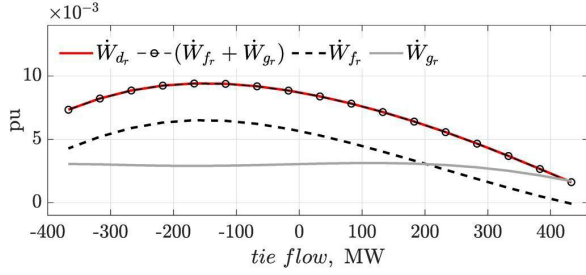
Our results on reconstructed charts from Section 5.1 showed that VLMs had near equal, and sometimes superior performance on tasks using the reconstructed charts compared to the originals. Figure 2 and Figure 7 show examples of this behaviour where GPT-5.4-mini correctly answers an identical question on the reconstruction, but not the original. To a human, the two charts appear nearly identical in content, with only minor stylistic differences. However, to a VLM these changes may matter. Low-level differences such as font choice, colors, and line thickness may contribute to VLM instability, but we also note there can be a more significant difference in the resolution of original and reconstructed charts. Original charts from the ChartQA, CharXiv, and ChartMuseum datasets can be low-resolution because they were scraped from a variety of sources. In contrast, our reconstructed charts are generated cleanly at high resolution from reverse-engineered code, which may benefit a VLM’s ability to parse information.

B.2 Conditional Variant Accuracy

Table 4 reports per-model CVA for each dataset. It complements Table 1 without repeating the OA, RA, VA, and RVC columns already reported in the main text. These values provide the model-level spread shown in Figure 3.

B.3 Update Diagnostics

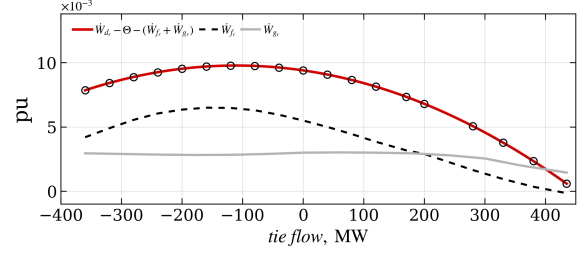
Table 5 reports per-model update outcomes on seed variants after original-chart success. Each row is computed with that model’s own diagnostic set



Question: At which tie flow value (MW) is the difference between $\dot{W}_{gr}$ and $\dot{W}_{fr}$ the smallest?

Ground Truth: 200 MW

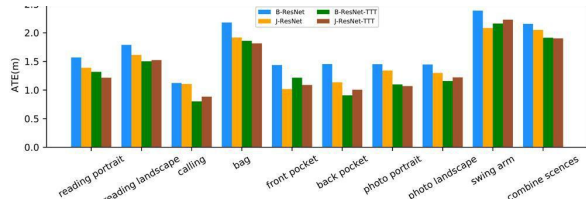
GPT-5.4-mini: The difference between $\dot{W}_{gr}$ and $\dot{W}_{fr}$ is the vertical gap between the gray solid curve and the black dashed curve. Scanning across the tie flow axis, that gap is smallest at the far right end of the plot, where the two curves nearly meet. **Answer:** About 440 MW



Question: At which tie flow value (MW) is the difference between $\dot{W}_{gr}$ and $\dot{W}_{fr}$ the smallest?

Ground Truth: 200 MW

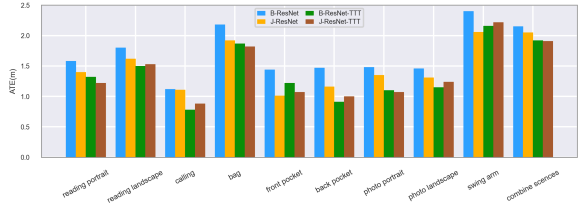
GPT-5.4-mini: The difference between $\dot{W}_{gr}$ and $\dot{W}_{fr}$ is smallest where the two curves are closest on the chart. Visually, they come closest around a tie flow of about 200 MW. **Answer:** 200 MW



Question: In which activity does the B-ResNet-TTT model outperform the J-ResNet model by the largest margin?

Ground Truth: front pocket

GPT-5.4-mini: To find where B-ResNet-TTT outperforms J-ResNet by the largest margin, compare the green bar against the orange bar for each activity. The biggest gap where the green bar is lower than the orange bar appears in both "calling" and "back pocket." **Answer:** calling and back pocket



Question: In which activity does the B-ResNet-TTT model outperform the J-ResNet model by the largest margin?

Ground Truth: front pocket

GPT-5.4-mini: Compare the B-ResNet-TTT bar to the J-ResNet bar for each activity and find where the gap is largest. The largest positive gap appears in front pocket. **Answer:** front pocket

Figure 7: Additional examples where GPT-5.4-mini gives a correct answer on the reconstruction (right), but not the original (left). The reconstruction has fewer artifacts and higher resolution because it was generated cleanly, not scraped from public sources. Flawed observations shown in red text.

$D_{m,d}$ for each dataset; the table therefore should be read as model-specific conditional rates rather than rates over a single shared pool of variants. The table expands the group-level CU, NU, and SP patterns summarized in Table 2.

B.4 ChartMuseum Reasoning-Type Analysis

Table 6 reports the per-model counterpart to Figure 4, using the original ChartMuseum reasoning-type annotations. It uses the same conditional framing as the main CVA result: variants are included only for families where the model answered the original chart correctly. Thus, the comparison focuses on generalizability after original success rather than static difficulty alone. It provides the model-level counterpart to the aggregate reasoning-type pattern in Figure 4.

B.5 ChartMuseum Failure Case Studies

We compare representative reasoning traces for the first two ChartMuseum rows in Figure 6. The first case illustrates stale prediction (SP) with evidence suggestive of memorization. The second illustrates noisy update (NU), where models move away from the original answer but still fail to ground the counterfactual variant correctly.

Case 1: Stale prediction and possible memorization. In the first ChartMuseum case, the gold answer changes, but the representative model responses keep the original prediction. The additional traces below follow the same pattern as the Claude Sonnet 4.6 traces in Figure 5: the counterfactual reasoning reuses duration claims that better describe the original chart. For GPT-5.4, the reuse is nearly word-for-word: both traces say that "Watching TV" spans the widest age range, starts

Model	ChartQA	CharXiv	ChartMuseum
<i>Proprietary models</i>			
Claude Haiku 4.5	0.924	0.690	0.513
Claude Sonnet 4.6	0.942	0.785	0.649
Gemini 2.5 Flash	0.938	0.738	0.623
Gemini 2.5 Pro	0.931	0.814	0.630
GPT-4.1	0.911	0.664	0.533
GPT-4.1 mini	0.915	0.706	0.584
GPT-4o	0.905	0.543	0.447
GPT-5.4	0.952	0.848	0.634
GPT-5.4 mini	0.945	0.782	0.600
Proprietary average	0.929	0.730	0.579
<i>Open-source models</i>			
Gemma 4 E4B IT	0.887	0.515	0.490
InternVL3-8B	0.893	0.428	0.355
LLaVA-OneVision-Qwen2-7B	0.726	0.285	0.348
Pixtral-12B	0.836	0.435	0.363
Qwen2.5-VL-7B-Instruct	0.844	0.522	0.372
Qwen3-VL-8B-Instruct	0.920	0.621	0.533
Open-source average	0.851	0.468	0.410

Table 4: Conditional variant accuracy (CVA) by dataset and model. CVA is variant accuracy restricted to families where the model answered the original chart correctly.

in early childhood, continues into adulthood, and is longer than “Video Games” or “Going to Concerts.” Gemini 2.5 Pro is less exact, but preserves the same original-chart explanation after the counterfactual variant changes the longest Time Waster. This supports the interpretation that the failure can reflect reliance on parametric knowledge of the original chart or QA pair, not only a local visual slip.

Case 2: Noisy update from incomplete visual-to-text grounding. In the second case, the gold answer changes from “May 2024” to “Dec 2024” where Figure 9 shows the reasoning trace for each model. GPT-5.4 updates correctly, while Claude Sonnet 4.6 and Gemini 2.5 Pro update to “Nov 2024,” producing noisy updates. Both incorrect responses locate the late-2024 peak region but anchor it to the nearby visible “Nov” tick instead of inferring the missing December label. Claude Sonnet 4.6 collapses a November–December localization to November, while Gemini 2.5 Pro explicitly traces the peak to the displayed “Nov” tick.

B.6 CharXiv Failure Case Studies

CharXiv does not provide reasoning-type annotations, so we group selected NU/SP case-study families using ChartMuseum-inspired visual task categories (Tang et al., 2025), with more specific labels when needed. The groups cover trajectory tracking and judgment, spatial visual comparison, symbol-guided visual comparison, value identification or thresholding, and other visually grounded cases.

Original Chart

Gemini 2.5 Pro. The chart uses a color-coded legend. “Time Wasters” are represented by red. The two red activities are Watching TV and Going to Concerts. Watching TV begins in early childhood, around age 6, and continues through adulthood past age 32. Going to Concerts starts much later, around age 18. Therefore, Watching TV spans the longest duration among Time Wasters. **Answer:** Watching TV.

GPT-5.4. The red bands represent the “Time Wasters.” Comparing the labeled red activities, “Watching TV” stretches across the widest age range, starting in early childhood and continuing into adulthood, longer than “Video Games” or “Going to Concerts.” **Answer:** Watching TV.

Counterfactual Variant

Gemini 2.5 Pro. The legend indicates that “Time Wasters” are represented by red. The red activities are Watching TV, Video Games, and Going to Concerts. Watching TV starts in early childhood, around age 4, and continues to the end of the chart at age 32. Video Games begins around age 7 and seems to end around age 21. Going to Concerts starts in the late teens, around age 17, and continues until about age 31. Comparing start and end points, Watching TV has the longest duration. **Answer:** Watching TV.

GPT-5.4. The “Time Wasters” category is shown in red. Comparing the red-labeled activities, “Watching TV” stretches across the widest age range, starting in early childhood and continuing into early adulthood, longer than “Video Games” or “Going to Concerts.” **Answer:** Watching TV.

Figure 8: Additional reasoning traces for the first case study in Figure 6. Highlighted spans show key reasoning points for the original chart (blue) and their incorrect reuse on the counterfactual variant (red).

Counterfactual Variant

GPT-5.4. The metals price index is shown by the red line. Scanning for its highest point, the peak occurs near the end of 2024, specifically in December 2024. **Answer:** December 2024.

Claude Sonnet 4.6. Looking at the red line (Metals), the metals line appears to reach its highest point around November–December 2024, where it climbs to approximately 113–114, the highest point the red line reaches throughout the chart period. **Answer:** November 2024.

Gemini 2.5 Pro. The legend indicates that the red line represents Metals. The question asks for the highest point of this red line. By visually scanning the line, its highest value is reached late in 2024. Tracing this peak down to the x-axis, it aligns with the “Nov” tick mark, after “Aug 2024” and before “Feb 2025.” **Answer:** Nov 2024.

Figure 9: Counterfactual variant responses for the metals price case. Highlighted spans mark the correct visual-to-text grounding (green) and the incorrect grounding to the nearby “Nov” tick (red).

For each family, Table 7 reports whether wrong counterfactual answers are mostly noisy updates (NU) or stale predictions (SP), computed over proprietary model-seed pairs after original-chart success. Figures 10–17 show one representative case per figure, including the original chart, one counterfactual seed, gold answers, representative model answers, and a short case note; correct answers are marked in green and incorrect answers in red.

Model	ChartQA			CharXiv			ChartMuseum		
	CU	NU	SP	CU	NU	SP	CU	NU	SP
<i>Proprietary models</i>									
Claude Haiku 4.5	0.924	0.074	0.003	0.690	0.263	0.046	0.513	0.413	0.073
Claude Sonnet 4.6	0.942	0.054	0.004	0.785	0.169	0.046	0.649	0.285	0.066
Gemini 2.5 Flash	0.938	0.059	0.004	0.738	0.211	0.051	0.623	0.279	0.098
Gemini 2.5 Pro	0.931	0.066	0.003	0.814	0.143	0.043	0.630	0.278	0.092
GPT-4.1	0.911	0.080	0.009	0.664	0.259	0.077	0.533	0.339	0.128
GPT-4.1 mini	0.915	0.078	0.007	0.706	0.210	0.084	0.584	0.355	0.061
GPT-4o	0.905	0.088	0.007	0.543	0.357	0.100	0.447	0.418	0.135
GPT-5.4	0.952	0.045	0.004	0.848	0.115	0.037	0.634	0.275	0.091
GPT-5.4 mini	0.945	0.048	0.008	0.782	0.149	0.069	0.600	0.300	0.100
Proprietary average	0.929	0.066	0.005	0.730	0.208	0.062	0.579	0.327	0.094
<i>Open-source models</i>									
Gemma 4 E4B IT	0.887	0.103	0.010	0.515	0.362	0.123	0.490	0.393	0.117
InternVL3-8B	0.893	0.067	0.040	0.428	0.362	0.211	0.355	0.463	0.182
LLaVA-OneVision-Qwen2-7B	0.726	0.235	0.039	0.285	0.568	0.147	0.348	0.381	0.270
Pixtral-12B	0.836	0.147	0.017	0.435	0.427	0.137	0.363	0.509	0.129
Qwen2.5-VL-7B-Instruct	0.844	0.129	0.026	0.522	0.357	0.122	0.372	0.474	0.154
Qwen3-VL-8B-Instruct	0.920	0.071	0.009	0.621	0.263	0.117	0.533	0.372	0.094
Open-source average	0.851	0.125	0.023	0.468	0.390	0.143	0.410	0.432	0.158

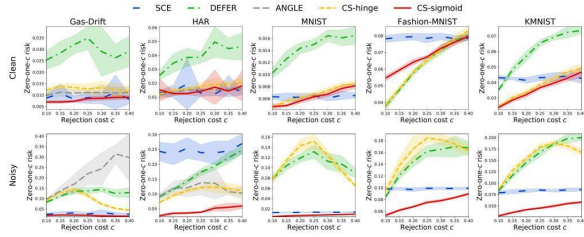
Table 5: Update outcome rates by dataset and model after original-chart success. Seed variants are decomposed into correct update (CU), noisy update (NU), and stale prediction (SP).

Model	Text	Visual/text	Synthesis	Visual
<i>Proprietary models</i>				
Claude Haiku 4.5	0.653	0.514	0.557	0.421
Claude Sonnet 4.6	0.930	0.695	0.486	0.524
Gemini 2.5 Flash	0.886	0.641	0.608	0.475
Gemini 2.5 Pro	0.952	0.615	0.664	0.468
GPT-4.1	0.738	0.521	0.463	0.448
GPT-4.1 mini	0.742	0.616	0.470	0.493
GPT-4o	0.583	0.591	0.367	0.310
GPT-5.4	0.852	0.705	0.583	0.511
GPT-5.4 mini	0.747	0.687	0.589	0.487
Proprietary average	0.787	0.620	0.532	0.460
<i>Open-source models</i>				
Gemma 4 E4B IT	0.670	0.600	0.443	0.344
InternVL3-8B	0.425	0.444	0.360	0.269
LLaVA-OneVision-Qwen2-7B	0.300	0.391	0.367	0.310
Pixtral-12B	0.383	0.420	0.400	0.328
Qwen2.5-VL-7B-Instruct	0.433	0.410	0.325	0.325
Qwen3-VL-8B-Instruct	0.680	0.521	0.583	0.411
Open-source average	0.482	0.464	0.413	0.331

Table 6: ChartMuseum CVA by reasoning type and model. CVA is conditioned on original-chart success. Reasoning-type names follow the ChartMuseum scheme.

Category	NU	SP	Total
Trajectory tracking and judgment	10	2	12
Value identification or thresholding	5	0	5
Spatial visual comparison	4	0	4
Symbol-guided visual comparison	3	1	4
Other visually grounded cases	1	1	2
Total	23	4	27

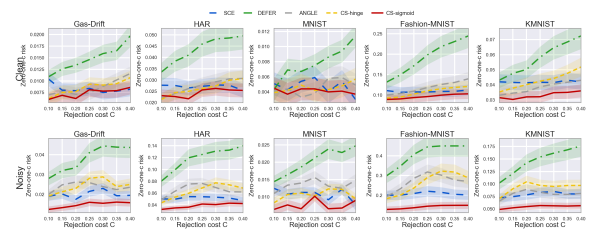
Table 7: CharXiv NU/SP case-study summary. Rows count selected families by visual task category; NU and SP indicate the dominant wrong-update mode over proprietary model-seed pairs after original-chart success.



Question: In the Noisy MNIST condition, which model has the greatest drop in zero-one risk at rejection cost $c = 0.35$?
Original. Gold: CS-hinge.
GPT-5.4: CS-hinge. Claude Sonnet 4.6: CS-hinge. Gemini 2.5 Pro: CS-hinge.

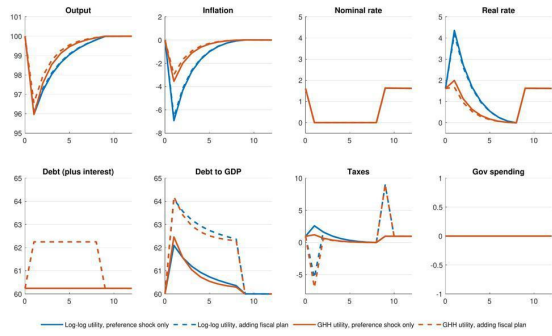
Update outcome rates: CU 0.40 / NU 0.60 / SP 0.00.

Case note: Trajectory tracking and judgment. The answer requires first locating the Noisy MNIST panel among several datasets, then reading several model curves at the same rejection cost. The target is not the lowest curve overall, but the largest local decrease around $c = 0.35$, so small mistakes in curve identity or local slope can change the answer.



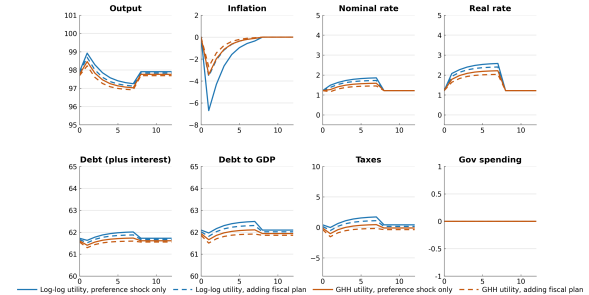
Question: In the Noisy MNIST condition, which model has the greatest drop in zero-one risk at rejection cost $c = 0.35$?
Counterfactual. Gold: SCE.
GPT-5.4: ANGLE. Claude Sonnet 4.6: CS-sigmoid. Gemini 2.5 Pro: CS-sigmoid.

Figure 10: CharXiv case study: trajectory tracking in risk curves.



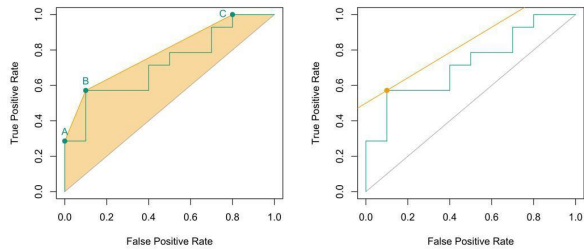
Question: Between $x = 0$ and $x = 5$, which subplot has curves with the most negative relation to the inflation subplot?
Original. Gold: Real rate.
GPT-5.4: Real rate. Claude Sonnet 4.6: Real rate. Gemini 2.5 Pro: Real rate.
Update outcome rates: CU 0.42 / NU 0.15 / SP 0.43.

Case note: Trajectory tracking and judgment. The question constrains the comparison to the interval $x = 0$ to $x = 5$, so the model must compare local relationships rather than the full-subplot trend. It also requires relating each candidate subplot to the inflation subplot, which makes stale answers plausible when the counterfactual changes which subplot is most negatively related.



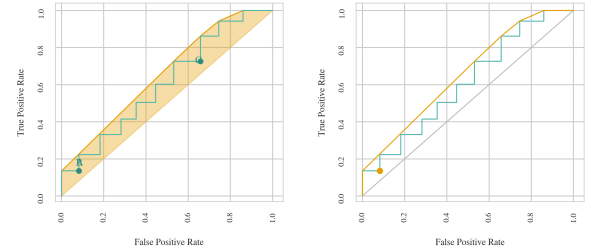
Question: Between $x = 0$ and $x = 5$, which subplot has curves with the most negative relation to the inflation subplot?
Counterfactual. Gold: Output.
GPT-5.4: Real rate. Claude Sonnet 4.6: Real rate. Gemini 2.5 Pro: Output.

Figure 11: CharXiv case study: trajectory comparison across economic subplots.



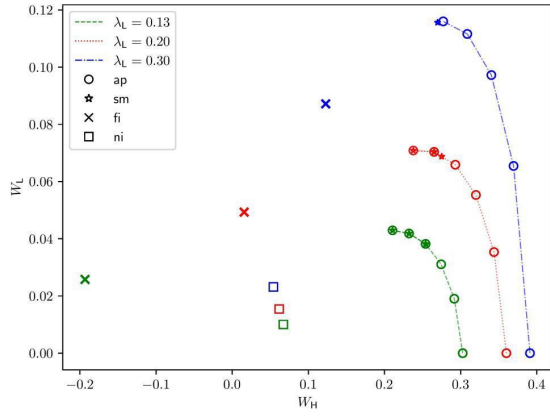
Question: How many steps separate point B and C in the figure on the left?
Original. Gold: 4.
GPT-5.4: 4. Gemini 2.5 Pro: 4. GPT-5.4 mini: 4.
Update outcome rates: CU 0.20 / NU 0.77 / SP 0.03.

Case note: Spatial visual comparison. The answer is a count over the discrete staircase path, not a continuous distance between points B and C. The model must identify the correct panel, follow the step sequence in order, and avoid substituting nearby axis values or the visually salient diagonal baseline for the path length.



Question: How many steps separate point B and C in the figure on the left?
Counterfactual. Gold: 6.
GPT-5.4: 10. Gemini 2.5 Pro: 4. GPT-5.4 mini: 5.

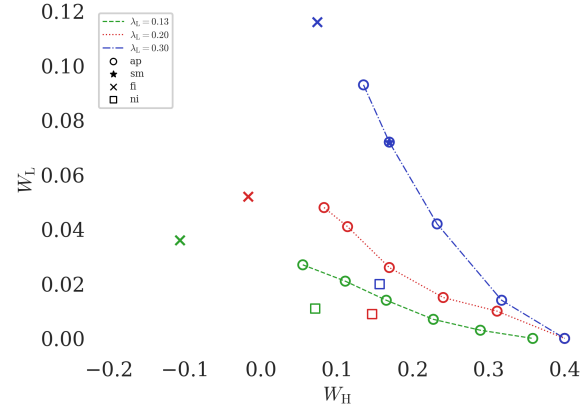
Figure 12: CharXiv case study: spatial step counting.



Question: Which line is furthest away from its f_i value from the W_H axis?
Original. Gold: $\lambda_L = 0.13$.
 GPT-5.4: $\lambda_L = 0.13$. Claude Sonnet 4.6: $\lambda_L = 0.13$. Gemini 2.5 Flash: $\lambda_L = 0.13$.

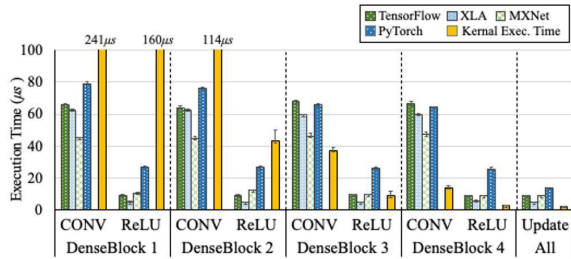
Update outcome rates: CU 0.30 / NU 0.70 / SP 0.00.

Case note: Spatial visual comparison. The answer depends on binding each colored trajectory to its corresponding f_i marker and comparing the marker's horizontal distance from the W_H axis. Because the curves, open markers, and f_i symbols are close together, a model can correctly recognize the chart type while still comparing the wrong visual object.



Question: Which line is furthest away from its f_i value from the W_H axis?
Counterfactual. Gold: $\lambda_L = 0.13$.
 GPT-5.4: $\lambda_L = 0.30$. Claude Sonnet 4.6: $\lambda_L = 0.30$. Gemini 2.5 Flash: $\lambda_L = 0.30$.

Figure 13: CharXiv case study: spatial line-marker binding.



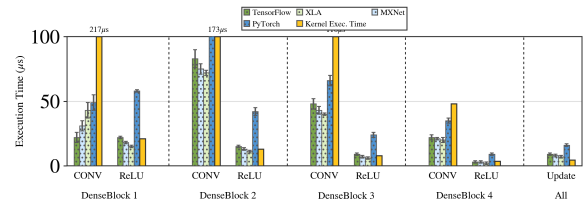
Question: Which library has the highest reduction in CONV execution time from DenseBlock 1 to DenseBlock 4?

Original. Gold: PyTorch.

GPT-5.4: PyTorch. Claude Sonnet 4.6: PyTorch. Gemini 2.5 Pro: PyTorch.

Update outcome rates: CU 0.34 / NU 0.60 / SP 0.06.

Case note: Symbol-guided visual comparison. The question asks for a reduction within one operation type, so the model must preserve the legend identity for each library and compare only the CONV bars from DenseBlock 1 to DenseBlock 4. ReLU bars, update bars, and the repeated library colors provide plausible distractors.

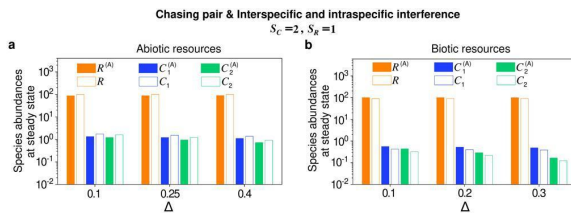


Question: Which library has the highest reduction in CONV execution time from DenseBlock 1 to DenseBlock 4?

Counterfactual. Gold: XLA.

GPT-5.4: MXNet. Claude Sonnet 4.6: PyTorch. Gemini 2.5 Pro: MXNet.

Figure 14: CharXiv case study: legend-bound bar comparison.



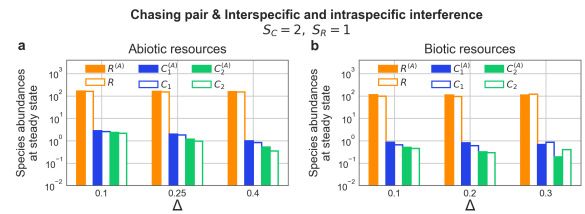
Question: Which subplot has each non-(A) data type always smaller than its (A) counterpart?

Original. Gold: Biotic resources.

GPT-5.4: Biotic resources. Claude Sonnet 4.6: Biotic resources. Gemini 2.5 Pro: Biotic resources.

Update outcome rates: CU 0.38 / NU 0.14 / SP 0.48.

Case note: Symbol-guided visual comparison. The answer is a universal comparison within a subplot: every non-(A) bar must be smaller than its matching (A) counterpart. This requires matching paired encodings across several categories, so a single violated pair should rule out a subplot even if its overall pattern looks similar.

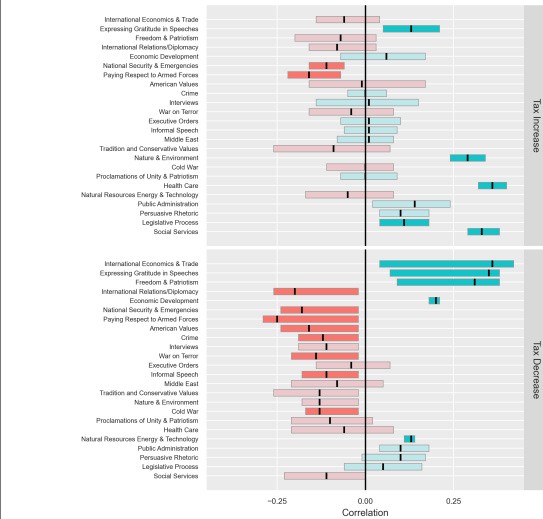
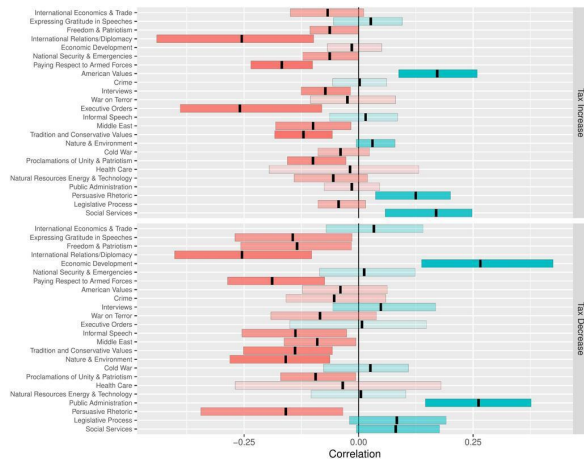


Question: Which subplot has each non-(A) data type always smaller than its (A) counterpart?

Counterfactual. Gold: Abiotic resources.

GPT-5.4: Abiotic resources. Claude Sonnet 4.6: Biotic resources. Gemini 2.5 Pro: Abiotic resources.

Figure 15: CharXiv case study: paired bar matching.



Question: How many bars exceed 0.25 in the Tax Decrease chart?

Original. Gold: 2.

GPT-5.4: 2. Gemini 2.5 Pro: 2. GPT-4o: 2.

Update outcome rates: CU 0.45 / NU 0.50 / SP 0.05.

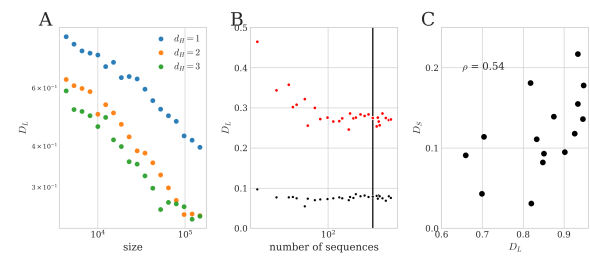
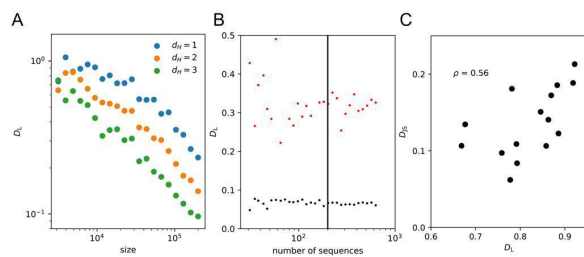
Case note: Value identification and thresholding. The question requires selecting the Tax Decrease facet, applying the 0.25 cutoff, and counting only bars that exceed that threshold. Nearby panels share the same visual scale, and several bars sit close to the cutoff, so both panel-binding and thresholding errors can change the count.

Question: How many bars exceed 0.25 in the Tax Decrease chart?

Counterfactual. Gold: 3.

GPT-5.4: 4. Gemini 2.5 Pro: 5. GPT-4o: 1.

Figure 16: CharXiv case study: thresholded bar counting.



Question: How many data points have a D_{JS} value less than 0.1?

Original. Gold: 3.

GPT-5.4: 3. GPT-5.4 mini: 3.

Update outcome rates: CU 0.50 / NU 0.50 / SP 0.00.

Case note: Value identification and thresholding. The model must identify the panel containing the D_{JS} values, apply a strict threshold at 0.1, and count individual points rather than infer from the overall trend or correlation annotation. Points near the cutoff make small value-reading errors sufficient to change the answer.

Question: How many data points have a D_{JS} value less than 0.1?

Counterfactual. Gold: 6.

GPT-5.4: 6. GPT-5.4 mini: 5.

Figure 17: CharXiv case study: thresholded scatter-point counting.